

Assignment No. 11

Steps to Run WordCount Program in Hadoop on Linux

1. Prerequisites:

- Ensure Hadoop is installed on the Linux machine.
- Java must be installed (Hadoop is built on Java).
- Ensure the HDFS and YARN services are running.

2. Set Up Hadoop Environment:

- Configure Hadoop by editing the configuration files such as `core-site.xml`, `hdfs-site.xml`, and `mapred-site.xml` located in the `/etc/hadoop/` directory. - Set the `JAVA_HOME` in the `hadoop-env.sh` file.

3. Start Hadoop Services:

- Start the Hadoop services by running the following commands:
```bash start-  
dfs.sh start-  
yarn.sh ```
- Use `jobs` to verify that all necessary services are running (e.g., NameNode, ResourceManager, DataNode, etc.).

#### 4. Prepare WordCount Code:

- Create a new Java file `WordCount.java`.
- The code should consist of a Mapper, Reducer, and a Main class for job submission.
- Ensure that you import the necessary Hadoop libraries.
- Code snippet for `WordCount.java`:

```
```java
import org.apache.hadoop.conf.Configuration; import
org.apache.hadoop.fs.Path; import
org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapreduce.Job; import
org.apache.hadoop.mapreduce.Mapper; import
org.apache.hadoop.mapreduce.Reducer; import
org.apache.hadoop.mapreduce.lib.input.FileInputFormat; import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat; import
java.io.IOException; import java.util.StringTokenizer; ```
```

5. Compile the Java Code:

- Compile the `WordCount.java` file by setting the Hadoop classpath:
```bash

Name:-Ganesh Malaji Ghule

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```
javac -classpath "`hadoop classpath`" -d classes WordCount.java
```
```

- This will generate the class files in the `classes/` directory.

6. Create a JAR File:

- Create a JAR file that includes all the compiled classes:

```
```bash jar -cvf wordcount.jar -C
classes/ .
```
```

7. Prepare Input Data:

- Place the input file (e.g., `input.txt`) in the HDFS. First, create a directory in HDFS:

```
```bash
hdfs dfs -mkdir -p /input
```
```

- Upload the input file to HDFS:

```
```bash
hdfs dfs -put /path/to/local/input.txt /input
```
```

8. Run the WordCount Program:

- Run the Hadoop job with the following command:

```
```bash
hadoop jar wordcount.jar WordCount /input /output
```
```

9. Handle Output Directory Issues:

- If the output directory already exists, delete it using the command:

```
```bash
hdfs dfs -rm -r /output
```
```

- Alternatively, you can choose a new output directory such as `/output\_new`.

10. View the Output:

- After the job completes, view the output file generated by the job:

```
```bash
hdfs dfs -cat /output/part-r-00000
```
```

11. Shut Down Hadoop Services:

- Once done, stop the Hadoop services:

```
```bash stop-
dfs.sh
stop-yarn.sh
```
```

Name:-Ganesh Malaji Ghule

Roll No:-37

By following these steps, you can run a basic WordCount program on a Hadoop cluster running on Linux. Make sure to configure Hadoop correctly, compile the Java code, create the JAR file, and upload the input to HDFS before executing the job.